

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Laws of Indices & Surds

Read the trigger, name the repeated structure, then use one clean line of algebra before calculating.

LAWS OF INDICES & SURDS · P1 · 01

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The Map

These notes are built from the Pure 1 topic booklet, its worked answers, and the WMA11 website solution bank. The whole topic reduces to five moves. Each move has a trigger, a first line, a short model, and a practice page.

01 POWER: Clean Powers

TRIGGER *simplify, divide powers, negative/fractional indices, form ax^n*

02 INDEX: Name The Target

TRIGGER *given $m = 2^n$, $y = a^x$, express in terms of the named letter*

03 SURD: Surd Clean-Up

TRIGGER *form $a\sqrt{b}$, collect like surds, expand surd brackets*

04 PAIR: Conjugate And Compare

TRIGGER *rationalise, $a + b\sqrt{k}$, linear equations with surds, exact geometry*

05 HIDE: Hidden Quadratic

TRIGGER *same exponential repeated, $y^{2/3}$, x^4 with x^2*

BANK EVIDENCE Local source: 36 questions across 26 problem pages and 55 answer pages. Website source: 13 checked WMA11 primary Indices & Surds questions, with step-by-step solution data.

FORMULA BANK

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad a^{-n} = \frac{1}{a^n}, \quad a^{m/n} = (\sqrt[n]{a})^m.$$

$$\sqrt{ab} = \sqrt{a} \sqrt{b}, \quad (u+v)(u-v) = u^2 - v^2, \quad \sqrt{a} + \sqrt{b} \neq \sqrt{a+b} \text{ in general.}$$

01 POWER: Clean Powers

TRIGGER *You see powers being multiplied, divided, raised again, or written in the form ax^n .*

BECOMES A bookkeeping question: coefficient with coefficient, exponent with exponent.

FIRST LINE TO WRITE

$$\sqrt{x} = x^{1/2}, \quad \frac{1}{x^n} = x^{-n}, \quad \frac{x^a x^b}{x^c} = x^{a+b-c}.$$

SIMPLEST STRATEGY

- 1 Put roots and denominators into power form.
- 2 Operate on exponents: add, subtract, or multiply.
- 3 Work coefficients separately from powers.
- 4 Express the result in the requested form.
- 5 Remove negative powers only if the question wants no negative indices.

WORKED MODEL

$$\frac{y^{5/7} \times y^{1/7}}{y^{3/7}} = y^{5/7+1/7-3/7} = y^{3/7}.$$

$$\frac{3y^3(2x^4)^3}{4x^2y^4} = \frac{3y^3(8x^{12})}{4x^2y^4} = \frac{6x^{10}}{y}.$$

HARD VARIANTS

- 1 If powers sit inside brackets, expand the power across every factor before collecting exponents.
- 2 If coefficients and powers are mixed, simplify coefficients separately and only then simplify the variable powers.
- 3 If the requested form bans negative or fractional powers, convert the final answer after the exponent work is complete.

BOTTOM LINE

Do the exponent arithmetic in one line. Most mistakes here come from moving the coefficient with the exponent.

SAME IDEA Clean every power before simplifying.

QUESTION

Simplify each expression.

$$(a) \quad \frac{x^4 x^{-7}}{x^{-2}}, \quad (b) \quad \left(\frac{1}{8} x^{3/2}\right)^{-4/3}, \quad (c) \quad \frac{5z^2(3z^{-1})^2}{15z^4}.$$

YOUR SOLUTION

02 INDEX: Name The Target

TRIGGER *The question says “given $m = 2^n$ ” or “given $y = a^x$, express in terms of m or y ”.*

BECOMES A replacement problem. Build the exact target power, then exchange it for the named letter.

FIRST LINE TO WRITE

$$\text{Target: } 2^n = m. \quad 2^{n+3} = 2^n \cdot 2^3.$$

SIMPLEST STRATEGY

- 1 Isolate the target power: 2^n , 3^x , or a^x .
- 2 New base: rewrite 16, 27, 81 as powers of the target base.
- 3 Distribute the exponent if it contains a sum or difference.
- 4 Exchange powers: $2^{12n} = (2^n)^{12}$.
- 5 Xchange the target for the named letter and simplify.

WORKED MODEL

Given $m = 2^n$, express in terms of m .

$$2^{n+3} = 2^n \cdot 2^3 = 8m.$$

$$16^{3n} = (2^4)^{3n} = 2^{12n} = (2^n)^{12} = m^{12}.$$

HARD VARIANTS

- 1 For bases like 16, $\frac{1}{8}$, $\sqrt{2}$, rewrite everything as powers of the named target base first.
- 2 If the target appears in a denominator or negative power, substitute first, then simplify the fraction.
- 3 If substitution creates an equation, solve it after the expression is in the named letter.

BOTTOM LINE

Name the target first. If the target appears, the substitution is almost finished.

03 SURD: Surd Clean-Up

TRIGGER *The answer must be in the form $a\sqrt{b}$, or the question asks you to simplify surds.*

BECOMES A common-surd question. Pull out square factors, then collect only matching surds.

FIRST LINE TO WRITE

$$\sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}.$$

SIMPLEST STRATEGY

- 1 Split each root using the largest square factor.
- 2 Unify terms by rewriting them with the same root where possible.
- 3 Reduce by collecting only like surds.
- 4 Deliver the final form, usually $a\sqrt{b}$.

WORKED MODEL

$$\begin{aligned} 4\sqrt{3} - \sqrt{12} + 4\sqrt{48} &= 4\sqrt{3} - 2\sqrt{3} + 16\sqrt{3} = 18\sqrt{3}. \\ (2\sqrt{2})^3 + 3\sqrt{2} &= 8(\sqrt{2})^3 + 3\sqrt{2} = 16\sqrt{2} + 3\sqrt{2} = 19\sqrt{2}. \end{aligned}$$

HARD VARIANTS

- 1 Expand surd brackets before collecting, especially $(a + b\sqrt{k})^2$.
- 2 Only like surds combine; $\sqrt{2}$ and $\sqrt{3}$ must stay separate.
- 3 For exact geometry or area answers, keep the surd exact until the final requested form.

BOTTOM LINE

A surd expression becomes easy only after every root is simplified.

04 PAIR: Conjugate And Compare

TRIGGER *There is a surd in a denominator, or the answer must be $a + b\sqrt{k}$, or a surd equation is linear in x .*

BECOMES A denominator-clearing question. Use the conjugate, then compare rational and surd parts if needed.

FIRST LINE TO WRITE

$$\frac{1}{a + \sqrt{b}} \times \frac{a - \sqrt{b}}{a - \sqrt{b}}.$$

SIMPLEST STRATEGY

- 1 Pick the matching partner: the surd itself or the conjugate.
- 2 Apply it to the numerator and denominator.
- 3 Identify the difference of squares in the denominator.
- 4 Reduce, then compare rational and surd parts if needed.

WORKED MODEL

$$\begin{aligned} \frac{14}{4 + \sqrt{2}} \times \frac{4 - \sqrt{2}}{4 - \sqrt{2}} &= \frac{56 - 14\sqrt{2}}{16 - 2} = 4 - \sqrt{2}. \\ x\sqrt{3} - 3 &= x + \sqrt{3} \Rightarrow x(\sqrt{3} - 1) = 3 + \sqrt{3}. \\ x &= \frac{3 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = 3 + 2\sqrt{3}. \end{aligned}$$

HARD VARIANTS

- 1 For $a + b\sqrt{k}$ comparison, expand fully and match the rational part and the surd part separately.
- 2 For denominators with two terms, use the conjugate every time; do not try to cancel a single surd term.
- 3 For surd equations, rationalise only after isolating the factor multiplying x .

BOTTOM LINE

The conjugate is not a trick; it is the fastest way to make a denominator rational.

05 HIDE: Hidden Quadratic

TRIGGER You see $9^x, 3^x, 3^{2x}$, or x^4, x^2 , or $y^{2/3}$ repeated.

BECOMES A quadratic in one repeated expression. Name the repeated expression, solve, reject invalid values, then return to the original variable.

FIRST LINE TO WRITE

$$p = 3^x \Rightarrow 9^x = p^2, \quad 3^{x+2} = 9p, \quad 3^{x-1} = \frac{p}{3}.$$

SIMPLEST STRATEGY

- 1 Highlight the repeated expression.
- 2 Introduce one new letter for it.
- 3 Do the quadratic in that new letter.
- 4 Exit back to the original variable, rejecting impossible values.

WORKED MODEL

Let $p = 3^x$. Then $9^x = p^2$, $3^{x+2} = 9p$, and $3^{x-1} = p/3$.

$$3p^2 + 9p = 1 + \frac{p}{3} \Rightarrow 9p^2 + 26p - 3 = 0$$

$$(9p - 1)(p + 3) = 0.$$

Since $p = 3^x > 0$, $p = \frac{1}{9}$. Therefore $3^x = 3^{-2}$, so $x = -2$.

HARD VARIANTS

- 1 For exponential quadratics, reject negative substitution values because $a^x > 0$.
- 2 For x^4, x^2 substitutions, return from $u = x^2$ by taking both square-root signs when $u > 0$.
- 3 For fractional powers such as $y^{2/3}$, name the repeated power and check that the returned values satisfy the original equation.

BOTTOM LINE

The substitution is the question. After that, it is just a quadratic.

SAME IDEA Name the repeated expression and return at the end.

QUESTION

$$(a) \quad 2(4^x) - 8(2^x) = 17(2^{x-1}) - 4.$$

Solve exactly.

$$(b) \quad x^4 - 3x^2 - 40 = 0.$$

Find the real values of x .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
$x^a x^b, \frac{x^a}{x^b}, (x^a)^b$	Write the operation above the expression: add for multiply, subtract for divide, multiply for a power.
$m = 2^n, y = a^x$	Write the target power first, then rebuild the expression around it.
$\sqrt{48}, \sqrt{75}, \text{form } a\sqrt{b}$	Pull out the largest square factor before collecting terms.
$\frac{1}{a+\sqrt{b}}, a+b\sqrt{k}$	Multiply by the conjugate and compare rational/surd parts.
9^x with 3^x , or x^4 with x^2	Let the repeated expression be one letter and solve a quadratic.

COMMON RESCUE LINES

$$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}, \quad \sqrt{a} - \sqrt{b} \neq \sqrt{a-b}.$$

$$(a + b\sqrt{5})^2 = a^2 + 5b^2 + 2ab\sqrt{5}.$$

$$Y = a^2 \Rightarrow a^4 - 49a^2 + 180 = 0 \text{ becomes } Y^2 - 49Y + 180 = 0.$$